Data Source Matrix

Fields	Primary Source	Format	Freshness
Fund Details	Static Config per product	JSON	Rare
AUM	Computed Daily	Derived	Daily
NAV	Computed Locally	Derived	Daily
Portfolio	Backend	Database	Api Call
Sector	Static Seed / From Portfolio metadata	File / JSON	As Portfolio Changes
Country/Region	Static Seed / From Portfolio metadata	File / JSON	As Portfolio Changes
Market Cap	NIFTY Data or other Index Data	JSON	Daily
Type	Static Seed / From Portfolio metadata	File / JSON	As Portfolio Changes
Risk	Static Seed / From Portfolio Metadata	File / JSON	As changes
Load Structure	Static Config per product	JSON	Rare
Quantitative Indicator	Computed Locally	Derived	Daily
Performance	Computed Locally	JSON	Daily
FX(Eg USD to INR)	yahoo finance	JSON	Daily
Daily Prices(per Security)	yahoo finance	JSON	Daily
Benchmark(NAV/Time)	yahoo finance	JSON	Daily

Asset Classification

Fund Details :
- Name
- Objective
- Type of Scheme
- Date of Inception (Launch Date)
- Fund Managers
- AUM (Assets Under Management)
- Average AUM for the Month or last year or day (Not needed it we update daily)
- NAV(Net Asset Value)
- Benchmark Index (eg NIFTY 500 TRI)
- Minimum Application / Additional Purchase Limit
- Minimum SIP Investment Amount
- Exchange Listed
- ISIN (International Securities Identification Number)

AUM(Assets Under Management) = NAV * Total number of Outstanding units

- Total marcket value of all financial assets
- It fluctuates when marcket goes up and down

How AUM changes in a factsheet ?
- the change in rising and falling values of securites in the fund
- net investor flows (fresh investments - redeptions)

Formula ??

AUM = Marcket value of Underlying assets + Inflow - Outflow

NAV(Net Asset Value) = Total Assets / Number of outstanding shares

- Determine exact buying and selling price

How NAV changes in a factsheet ??
- changes with the holdings value
- any intrest earned / dividend received from holdings , increases assets value , increases NAV value
- When a fund pays out to the investors , its total asset value decreases , so NAV value decreases .

Formula ??

NAV = (Total Assets - Total liabilities) / Total outstanding units

Porfolio : a curated collection of assets and investments

Types of holdings ??
- Stocks(Equity)
-Debt(Bonds)
-Cash
-REITs / InvITs

How it changes in Factsheet ??
- When price of a stock increases then the percentage of total assets that stock increases
- Fundamanager actively buys and sells some assets to maintain cashflow and rebalancing of portfolio

Factor Determining ??
- Fund Manager
- Market

Asset Classification : determined by the fund house and fund manager and regulated by the regulatory authority(SEBI)

How can assests and be classified ??
- Sector
- Country/Region
- Marcket Cap
- Equity / Debt / Cash / REITs/InvITs etc
- Based on Risk

Load Structure : fee or penalty charged by AMC(Asset management Company)

Types ??
- Entry Load : Fee as set by the AMC
- Exit Load : Penalty charged by AMC if investor withdraws in a particular time

Sector : classification on the basis of sector

Types ??
- Finance
- IT
- Oil & Gas / Energy
- Automobile and components
- FMCG
- Telecommunication
- Metal and Mining , etc

Country / Region : classification based on region

Types??
- Indian / Japan / UK / USA / China etc
- Asia-Pacific / North America / Africa / Europe etc

Marcket Cap : classification based on marcket cap

Types ??
- Large Cap
- Mid Cap
- Small Cap

Equity / Debt /Cash / REITs / InvITs :

Based on Risk : classification based on Risk

Types ??
- Very Low
- Low
- Moderate
- High
- Very High

Quantitative Indicator : are statistical metric that measure a fund's historical risk, volatility, and return efficiency

Types ??
- Volatility and Market Sensitivity
- Risk adjusted return indicators
- Manager performance Indicator

Standard Deviation : measures the volatility or price swing of a fund (High SD means highly volatile fund)

Formula ??

$$\sigma = \sqrt{\frac{\sum (R_i - \bar{R})^2}{n-1}}$$

- R_i: The return of the fund in period i (e.g., month i).
- $\bar{R}$: The mean (average) return of the fund over the entire period.
- n: The total number of periods (e.g., 36 months).
- Note: Factsheets annualize this monthly figure by multiplying it by $\sqrt{12}$.

Beta : Measures fund's sensitivity especially relative to the standard ?

Formula ??

$$\beta = \frac{\text{Covariance}(R_p, R_m)}{\text{Variance}(R_m)} = \frac{\text{Cov}(R_p, R_m)}{\sigma_m^2}$$

- R_p: Return of the portfolio (fund).
- R_m: Return of the market benchmark index.
- $\text{Cov}(R_p, R_m)$: How the fund and market move together.
- σ_m^2: The variance of the market benchmark's returns.

Sharpe Ratio : Excess return generated by unit of total risk

Formula ??

$$\text{Sharpe Ratio} = \frac{R_p - R_f}{\sigma_p}$$

- R_p: Expected or average annualized return of the portfolio.
- R_f: Risk-free rate of return (typically the yield of the 91-day Government Treasury Bill).
- σ_p: Annualized Standard Deviation of the portfolio.

Treynor Ratio : Similar to Sharpe Ration , but calculate against market risk (Beta)

Formula ??

$$\text{Treynor Ratio} = \frac{R_p - R_f}{\beta_p}$$

- R_p: Average annualized return of the portfolio.
- R_f: Risk-free rate of return.
- β_p: The Beta of the portfolio.

Alpha : Represents the value that a fund manager adds or subtracts relative to the benchmark's return

Formula ??

$$\alpha = R_p - [R_f + \beta_p(R_m - R_f)]$$

- R_p: Actual annualized return of the portfolio.
- R_f: Risk-free rate of return.
- β_p: Beta of the portfolio.
- R_m: Actual annualized return of the benchmark index.
- $[R_f + \beta_p(R_m - R_f)]$: The return the fund should have achieved based on its risk level.

R-Squared : Measures the percentage of a fund's movements that can be explained by movements in its benchmark index

Formula ??

$$R^2 = (r)^2 = \left(\frac{\text{Cov}(R_p, R_m)}{\sigma_p \sigma_m}\right)^2$$

- r: Pearson correlation coefficient between the fund and the benchmark.
- $\text{Cov}(R_p, R_m)$: Covariance of the fund and benchmark returns.
- σ_p: Standard deviation of the portfolio.
- σ_m: Standard deviation of the benchmark index.

Performance : critical part shows that how much money the fund has made for its investors

Types and Categories ??
- Key Metrics
- Tri-Comparison Framework
- Evaluating performance and tracking error

- CAGR Returns (Compound Anual Growth Rate) : For eg . over 5 years the fund grew with an average of 15%
- PTP Return (Point to Point) : If you had invested ₹10,000 at the inception of this fund, it would now be worth ₹31,308.
- TRI (Total Return index) : This means the benchmark returns include both stock price changes and the dividends re-invested, which is the most accurate way to measure performance.

- The Fund: SBI Nifty 50 ETF
- Scheme Benchmark (Primary): Nifty 50 TRI
- Additional Benchmark (Secondary): BSE Sensex TRI

- 1-Year Horizon: The benchmark dropped -0.28%, and the fund dropped -0.32%. The minute difference of 0.04% is the tracking error, which is exceptionally low and indicates great fund management.
- 3-Year Horizon: The benchmark returned 11.78%, while the fund tracked right behind at 11.71% (a minor lag of 0.07%).
- 5-Year Horizon: The benchmark returned 11.69%, and the fund delivered 11.61%.

Decision 01 - Storage ??

Options ??
1. Database
2. Database + Redis

Approach :

Database + Redis , we will store the data in redis completly till the pipeline ends

Decision 03 - Holdings data source ?

Approach :

The assignment does not provide any source of holdings , so we will use databse for it and create a api to manage that , can be done only with particular user .

Decision 04 - Where does prices/FX/classification comes from ??

Options :
1. yfinance api
2. dual-source (yfinance + backup 1)
3. multisource (yfinance + backup 1 + backup 2)

Approach :

The assignment explicitly told to use yfinance api . For now we will stick to the yfinance api , but for production we must have backups so the we do not completly depends on a single source only .

Decision 05- how we handle classification of holdings ??

Options :
1. form database (same as holdings)
2. yfinance
3. yfinance + database

Approach :

We will pull the classification from yfinance and does not change daily , so we will pull daily and update in the database .

Decision 06- NAV??

Approach :

We will compute NAV daily and store historical data for graphs .

Decision 07- pipeline run time ??

Approach :

Can be configurable in .env and manual run option will be there .

Decision 08- Database ??

Options :
1. NoSQL
2. SQL

Approach :

Data is aggressively relational. Transactions matter for atomic snapshot writes. SQL GROUP BY is the natural query for exposures. Mongo's $lookup is painful at this depth.

Decision 09 - How do we store decimal financial values?

Approach :

NUMERIC(20,6) for prices, NUMERIC(10,8) for weights, NUMERIC(20,2) for money

Decision 10- Job schedular choice ??

Options :
1. APScheduler
2. celery + Redis
3. host corn jobs
4. Airflow

Approach :

We will use APScheduler , celery is overkiller , corn jobs we need to look for the logs when the fetch fails .

Decision 11- How do we guarantee users never see a half-updated factsheet?

Approach :

Each pipeline run wraps all canonical writes in one transaction. Rollback on any failure. Never show users a partial snapshot. Either the day's factsheet is fully written, or yesterday's stays live.

Decision 12- How do we minimise pipeline run time API calls ??

Approach :

We will try to call and collect all the data in 1 or 2 yfinance api calls . Also we will track the time taken by each process . So that when the failure occures we can show the previous data to the user , no user side failures and run the pipeline again .

Decision 13 - Push full factsheet , or only required data through different urls?

Approach :

Only required data through different urls

Decision 14- What we will serve when the latest pipeline is still ongoing ??

Approach :

We will store the latest run pipeline data in redis cache , if the latest pipeline status is in progress , the user will always gets data form the database not redis . And when the pipeline ends , it will store all the data at once in Database in one atomic transaction .

Decision 15 - intraday worker works only after market hours or 24/7 ??

Approach :

US and Indian Market hours are different , we need to configure the time in .env